UNIVERSITY OF THE PHILIPPINES

Master of Science in Materials Science and Engineering

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Investigation of a-Si/SiO_2 distributed Bragg reflectors with InGaAs terahertz emitters

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CHAPTER 1

INTRODUCTION

Recently, there has been an interest in developing terahertz (THz) photoconductive antenna (PCA) that can be activated by lasers operating at telecommunication wavelengths of 1.55 µm due to the possibility of integration with existing technology[1]. These PCAs are used in THz time domain spectroscopy (TDS) for various applications, including biology[2], imaging[3], and pharmaceuticals[4], among others. A PCA consists of a semiconductor material where the terahertz generation or absorption occurs and two metal electrodes on the semiconductor surface separated by a few microns that forms the photoconductive gap[5]. When a femtosecond pulse from the laser illuminates the gap, carriers are generated in the semiconductor and swept by an electric field from the bias applied to the metal electrodes, and the movement of the carriers generates the terahertz wave.

One of the common semiconductor materials of choice for a PCA operating at the 1.55 µm wavelength is low temperature (LT) grown indium gallium arsenide (InGaAs), which has a bandgap of $E_g < 0.8\,\mathrm{eV}$ and can be grown lattice matched to an indium phosphide (InP) substrate via molecular beam epitaxy or molecular beam epitaxy (MBE)[6]. However, when an InGaAs PCA is illuminated by a laser there are losses involved due to the high refractive index of the material[7]. Adding an antireflective coating to the InGaAs PCA can increase the efficiency of the device by reducing the amount of light reflected by the film. The challenge lies in the design of a suitable antireflective coating for InGaAs.

1.1 Significance of the study

The development of THz PCAs compatible with telecommunication wavelengths holds great potential for integrating THz technology with existing optical systems. By enhancing the efficiency of THz generation, this study contributes to the advancement of THz TDS systems, which have diverse applications in fields such as biology, imaging, and pharmaceuticals. The current use of InP substrates for MBE growth of LT-InGaAs based PCAs allows creation of films operating at 1.55 µm wavelengths with acceptable strain. However, this setup faces limitations, including THz absorption by the InP substrate and inefficient laser energy absorption by the InGaAs thin film. By integrating a distributed Bragg reflector (DBR) stack, the study aims to address these problems, improving the THz intensity generated by the PCA and its overall performance. The findings could pave the way for more efficient THz devices, enhancing capabilities in scientific research, medical diagnostics, and industrial processes.

1.2 Objectives of the study

The main objective of this study is to enhance the intensity of THz waves generated by LT-InGaAs PCAs using an anti-reflection coating (ARC) stack made of amorphous silicon (a-Si)/SiO_2. Specifically, this will require the following objectives:

- to simulate an a-Si/SiO_2 DBR stack and fabricate it using electron beam deposition,
- to integrate the fabricated ARC stack with an InGaAs PCA, and
- to evaluate the THz intensity of the integrated LT-InGaAs a-Si/SiO_2 ARC PCA device using THz TDS.

1.3 Scope and limitations of the study

The study will only cover the integration of an LT-InGaAs PCA with an a-Si and SiO_2 ARC. As such, the study will not investigate the growth of the LT-InGaAs film using MBE and will also be limited to electron beam evaporation as the growth method for the a-Si/SiO_2. The characterization is also limited to the evaluation of the THz intensity of the integrated device and will not include applications of the developed system.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Terahertz photoconductive antenna

The range of frequencies between microwave and infrared in the electromagnetic spectrum is known as the THz range. It was also called the THz gap in the past due to the difficulty of finding emitters and detectors which worked in this frequency range[8]. THz radiation is non-ionizing and can penetrate through organic material such as plastic and clothing, thus it has potential use in biological characterization[2], detection of contraband materials[9], and chemical identification[10]. Some substances have specific THz absorption spectra which can be used to identify their composition, as shown in Figure 2.1. The interaction amongst the bonds and forces in the molecules result in vibrational modes visible in the THz region[9]. On the other hand, THz reflection can be used to detect concealed objects when the concealing material is transparent to THz radiation as in refthrough, as the case for dielectrics such as clothing[11]. A common setup for some of these applications involves THz TDS.

In THz TDS, a THz pulse is generated using a PCA. The shape of the electrodes influences the pattern of the THz radiation generated by the PCA. Dipole and bowtie

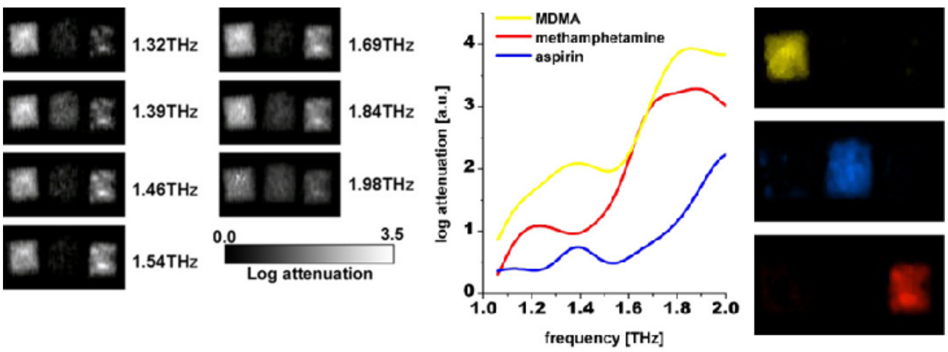

Figure 2.1: THz identification of MDMA, methamphetamine, and aspirin through a paper envelope. The substances are differentiated using their THz spectra[9]

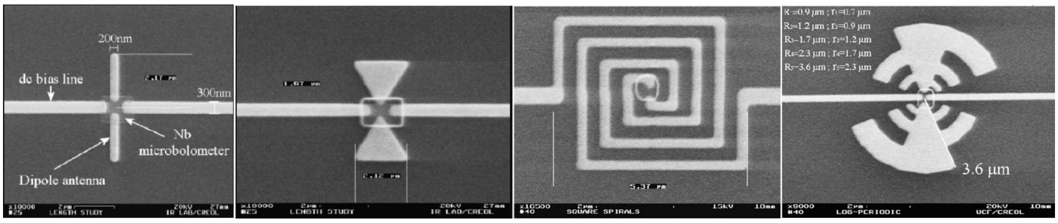

Figure 2.2: Commonly used THz PCA geometries; from left to right: dipole, bowtie, spiral, and log-spiral[12]

patterns demonstrate higher polarization ratios compared to spiral antennas while spiral and log-spiral antennas have a higher calculated radiation efficiency compared to bowtie and dipole antennas[12]. The shapes of the antennas are presented in Figure 2.2.

The emission mechanism for the PCA is as follows: the photoconductive material between the electrodes is illuminated with a femtosecond laser pulse, which generates photocarriers that are swept by an electric field applied to the electrodes of the PCA[5], [6]. The transient current $i_{PC}\left(t\right)$ resulting from the moving carriers generates electromagnetic radiation according to Maxwell's equations, with the intensity of the electric field component of the THz wave radiated by a dipole antenna expressed byEquation 2.1

$$E_{THz}\left(r,\theta,t\right)=\frac{l_e}{4\pi\varepsilon c^2 r}\frac{\partial i_{PC}\left(t\right)}{\partial t}\sin\left(\theta\right) \tag{2.1}$$

where l_e is the antenna length, ε the permittivity, c the speed of light, r the distance from the dipole, and θ the angle relative to the dipole axis[5].

The transient current density term $i_{PC}\left(t\right)$ is proportional to the convolution integral of the carrier density $n\left(t\right)$ and electron velocity $v_e\left(t\right)$ as shown in Equation 2.2[5].

$$i_{PC}\left(t\right)\propto P_{pulse}\left(t\right)\otimes\left(n\left(t\right)v_e\left(t\right)\right). \tag{2.2}$$

The carrier density is described by

$$\frac{dn}{dt}=\eta\frac{P_{\text{pulse}}(t)}{h\nu}-\frac{n}{\tau_c} \tag{2.3}$$

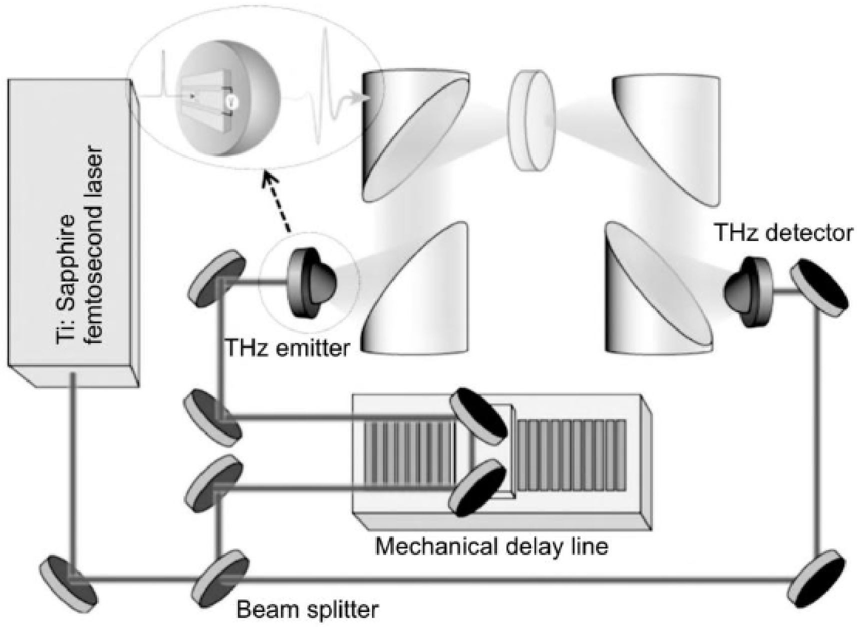

Figure 2.3: Schematic of a THz TDS setup with a femtosecond laser and two photoconductive antennas[5]

and using a 1D Drude-Lorentz model, the electron velocity by

$$\frac{dv_e}{dt} = -\frac{v_e}{\tau_s} + \frac{e}{m^*}E \tag{2.4}$$

where η is the absorption efficiency, $P_{\text{pulse}}(t)$ the ultrashort laser pulse power considered over time, τ_c, τ_c, and m^* the carrier lifetime, momentum relaxation time, electron effective mass respectively, and E the local electric field[5].

A photoconductive antenna is also used to detect THz radiation, with the incoming THz pulse focused together with a femtosecond pulse split from the same source at the gap of the antenna. As the femtosecond probe pulse arrives at the gap, carriers are excited and then accelerated proportionally by the electric field provided by the incoming THz pulse[6]. The difference of the probe pulse and the THz pulse provides a measurement in the time domain for the transient of the THz pulse. A schematic of a complete THz TDS system with emitter and detector is shown in Figure 2.3.

2.2 Low temperature grown InGaAs

From the previously discussed equations, the performance of the THz TDS system depends on multiple factors such as the properties of the femtosecond laser pulse and the quality of the photoconductive material. Modification of the photoconductive material is done to achieve short electron-hole recombination times, high carrier mobilities, and high breakdown electric fields, which all contribute to an increase in the intensity of generated THz radiation[13].

Some of the photoconductive materials used for THz TDS systems include LT grown GaAs and InGaAs[14], [15], [16]. LT-GaAs is grown via molecular beam epitaxy at $200\,^{\circ}\mathrm{C}$ to $300\,^{\circ}\mathrm{C}$ instead of the usual growth temperature around $600\,^{\circ}\mathrm{C}$[17]. The growth of GaAs and related compounds at relatively low temperatures and slower rates in molecular beam epitaxy compared to other epitaxial techniques result in mechanisms which are dominated by surface kinetics and surface chemistry[18]. MBE growth can be divided into two stages: the initial stage involving processes such as the chemisorption of molecules, their migration across the surface, and bond formation; and the subsequent stage which involves thermodynamic interaction and redistribution in the surface regions to form the crystal[18]. The molecular beam which supplies the material is produced by evaporation or sublimation of high-purity solid sources at elevated temperatures in ultra-high vacuum, allowing the molecules to reach the substrate without interacting. When the molecules arrive at the substrate, they are chemisorbed to the surface and can migrate to energetically favorable step edges or nucleated islands and form bonds, gather at contaminations to form defects or agglomerate with similar species, or desorb from the surface as shown in Figure 2.4[19], [20]. The thermodynamic interaction and redistribution of the layers close to the surface to form the final crystal configuration has been observed during the growth process of InGaAs using Auger electron and x-ray photoemission spectroscopy[21], [22].

The low growth temperature coupled with excess arsenic during the process creates a non-stoichiometric material with a high concentration of point defects in the form

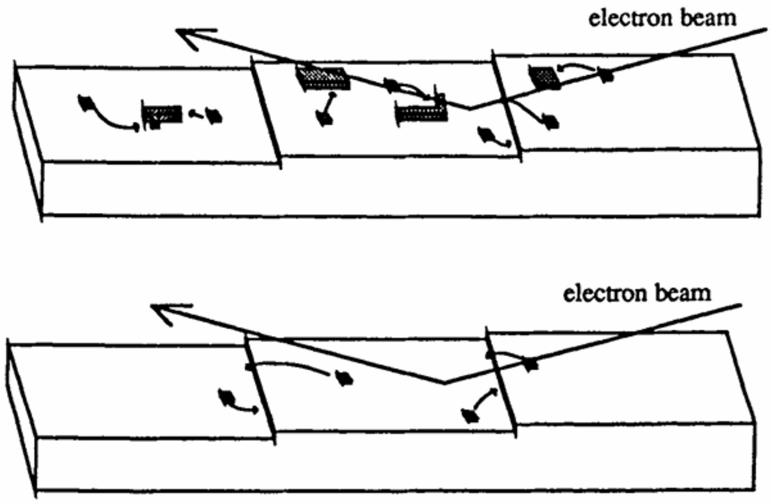

Figure 2.4: Kinetics of epitaxial growth observed using an electron beam. Molecules can form islands, migrate to step edges, or desorb from the surface.[18]

of As antisites As_{Ga} close to the center of the bandgap as shown in Figure 2.5 [23]. The As_{Ga} point defects can be found in the neutral As^0_{Ga} and charged As^+_{Ga} states, with reported concentrations of $1 \times 10^{20}/\mathrm{cm}^3$ and $1 \times 10^{10}/\mathrm{cm}^3$ respectively at a growth temperature of $200\,°\mathrm{C}$[24]. Close to the valence band, acceptor vacancies V_{Ga} form at equal concentration to As^+_{Ga}. As^+_{Ga} acts as an electron trap, and together with the other mid-bandgap states serve as a ready source of electrons that can be easily promoted to the conduction band, which results in a much larger carrier concentration in the conduction band compared to ordinary material which relies solely on band-to-band transitions[25]. Despite the high defect concentration of LT-GaAs, the material has been reported to have high carrier mobility of around $200\,\mathrm{cm}^2/(\mathrm{V\,s})$[25].

The partial substitution of Ga atoms by In in the GaAs structure results in the ternary alloy $In_xGa_{1-x}As$. Controlling the bandgap and consequently absorption wavelength for InGaAs is done by changing the mole fractions of In and Ga as shown in Figure 2.6 [26]. The ratios of In and Ga are easily controlled by changing the fluxes of the corresponding sources during epitaxial growth.

A phase diagram for the surface reconstruction of $In_xGa_{(1-x)}As$ as a function of In composition on GaAs(100) and InAs(100) substrates is shown in Figure 2.7[27]. The gaps in the phase diagram indicate locations where the morphology of the surface is no longer uniform and flat enough to obtain data from scanning tunneling microscopy

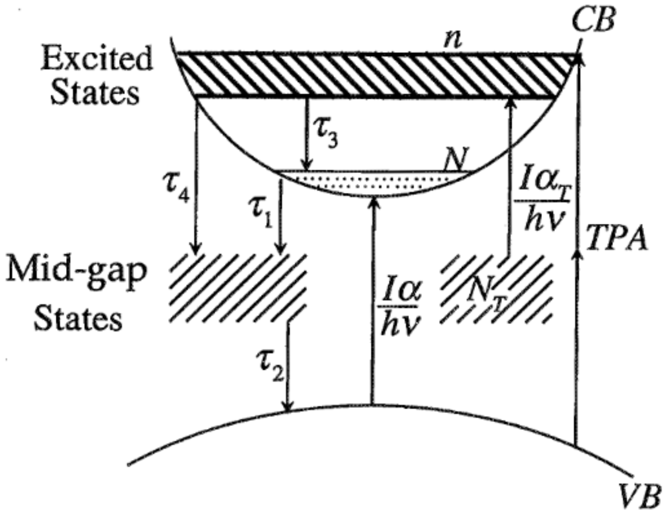

Figure 2.5: Energy level diagram of the defect states in LT-GaAs[25]

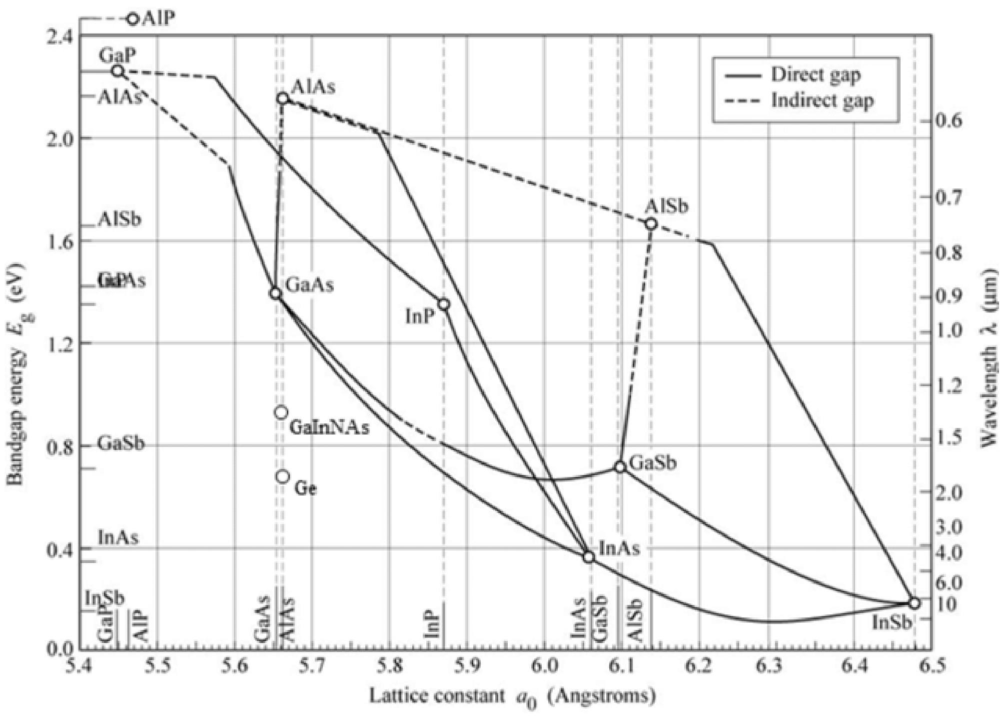

Figure 2.6: Bandgap versus lattice constant for various III-V compounds[26]

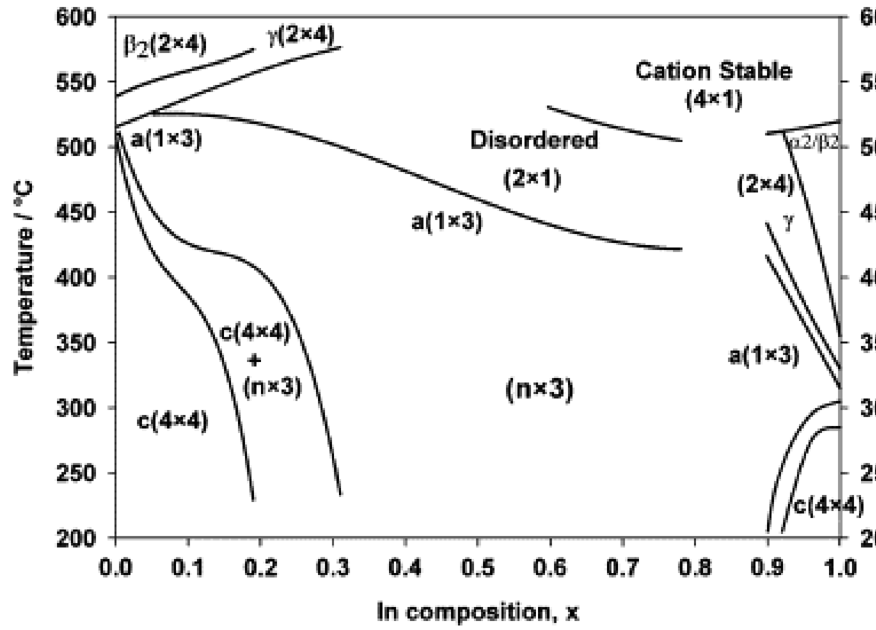

Figure 2.7: Phase diagram for the surface reconstruction of $In_xGa_{(1-x)}As$ using RHEED

(STM) and reflective high energy electron diffraction (RHEED).

For InGaAs on GaAs(100) the system is compressively strained and becomes progressively rougher as the In fraction increases, while InGaAs grown on InAs(100) is tensile strained, thus the choice of substrate is also important during epitaxy. Based on Figure 2.6 InP is the most suitable substrate lattice matched to InGaAs operating at $1.55\,\mu\mathrm{m}$, however InP is highly absorbing for THz radiation and has a relatively high refractive index which hinders efficient emission of THz radiation as shown in Figure 2.8[28].

2.3 Distributed Bragg Reflector

Increasing the absorption of the photoconductive material can be done by reflecting unabsorbed light from the laser back into the material by means of a distributed Bragg reflector[29]. A DBR is composed of a stack of alternating layers of materials having different refractive indices. When an electromagnetic wave propagates across the boundary of two materials with differing refractive indices, partial reflection of the field occurs if the refractive index of the succeeding material is greater than that of the material before it. If the thickness of each layer is set to a quarter of the wavelength of

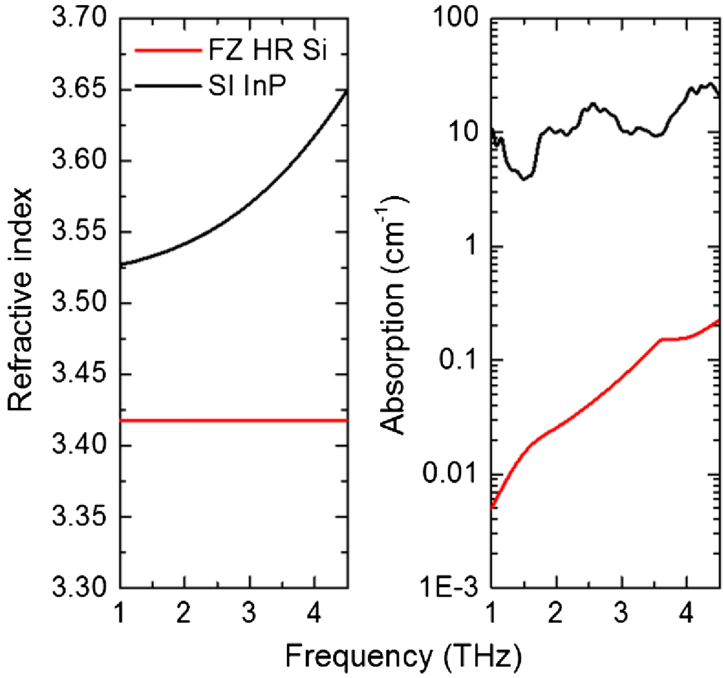

Figure 2.8: Refractive index and absorption coefficient of semi-insulating InP and float zone high resistivity Si in the THz range[5]

interest, the reflection from each successive interface is shifted by half the wavelength and the collection of all reflected components interfere constructively[30].

An explanation of the distributed Bragg reflector can be obtained from the wave equation as given by

$$\frac{d^2E}{dz^2} + \left\{ \frac{\omega^2}{c^2}[n_0 + n_1 \cos(2\beta_m z)] + j\frac{\omega}{c}[-n_0 i\gamma_0 + \gamma_1 \cos(2\beta_m z + \theta)] \right\} E = 0 \quad (2.5)$$

where the involved terms are $\beta_0 = \frac{\omega}{c} n_0$ the phase constant of the unperturbed mode, $g_0 = \gamma_0/2$ the field coefficient, $g_1 = \gamma_1/2$ the perturbation in the field coefficient, $\kappa = \frac{\omega}{c} \cdot \frac{n_1}{2} + j\frac{g_1}{2}$ the coupling coefficient, and $2\beta_m = \frac{2\pi}{\Lambda}$ where Λ is the mechanical periodicity $\lambda_B/2n_0$. By using some approximations,Equation 2.5 can be reduced to the matrix equation

$$\begin{bmatrix} f'' \\ r'' \end{bmatrix} - p^2 \begin{bmatrix} f \\ r \end{bmatrix} = 0 \quad (2.6)$$

where $p^2 = \kappa^2 + (g_0 - j\delta)^2$. Returning to the example of the propagating electromagnetic wave E_+ at $z = -L/2$, if the medium through which it propagates is periodic

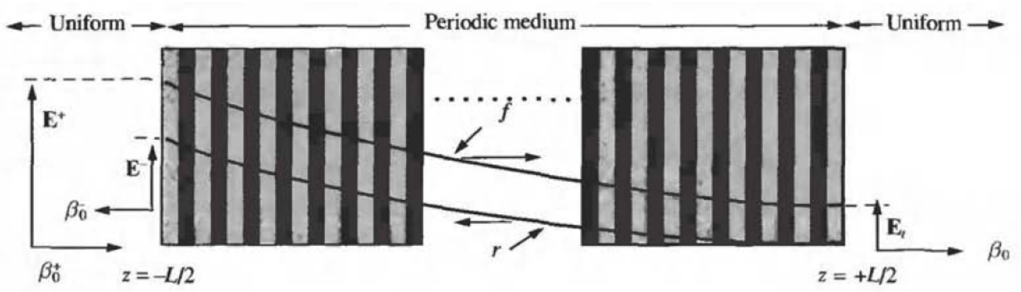

Figure 2.9: Geometry of a distributed Bragg reflector medium with a source of electromagnetic waves located on the left. The amplitude of $\mathbf{E}$ decreases as it propagates through the structure[31].

as shown in Figure 2.9, then a reverse wave r is generated which contributes to the reflected wave E_{-} at the same boundary. A portion of the propagating wave may reach the right boundary of the periodic structure at $z = L/2$ and turn into a transmitted wave E_t. There is no reverse wave at the right boundary since the medium beyond does not have the properties to reflect the wave. When the field gain coefficient $g_0 = 0$ (and also for g_1), the coupling coefficient κ is a real number, andEquation 2.6 simplifies to the equation

$$(pL)^2 = (\kappa L)^2 - (\delta L)^2. \tag{2.7}$$

The power transmission and reflection coefficients at the band center are given by

$$\begin{aligned} R &= |S_{11}|^2 \approx [1 - 2e^{-2\kappa L}]^2 \\ T &= |S_{12}|^2 \approx 4e^{-2\kappa L} \end{aligned}. \tag{2.8}$$

The region at $-\kappa L < [\delta L = (\beta_0 - \beta_m)L] < \kappa L$ is known as the stop band of the DBR, since the wavelengths which satisfy this condition are heavily attenuated while propagating through the structure, as shown in the power transmission graph in Figure 2.10. The amplitudes of the propagating electric field decrease exponentially as shown in Figure 2.9. Some of the methods used to simulate the operation of DBRs include finite element analysis [32] and the transfer matrix method[33].

There are multiple material pairs that can be chosen for the DBR A commonly used material pair is AlAs/GaAs, which can be grown via MBE[34]. Other material

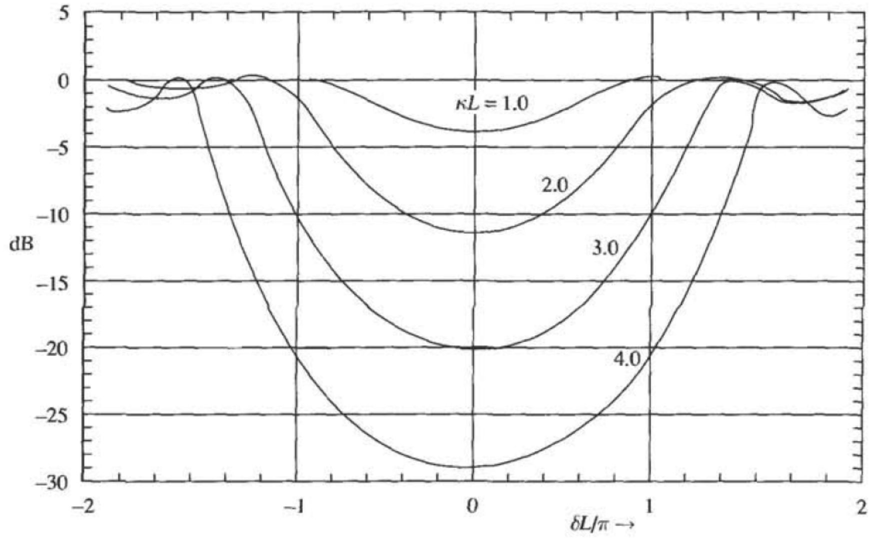

Figure 2.10: Relative power transmission of a DBR versus the detuning parameter $\delta L/\pi$ for various coupling coefficient κL values [5]

systems which have been grown epitaxially for vertical cavity surface emitting lasers based on InP substrates operating at $1.55\,\mu\mathrm{m}$ include InGaAsP/InP, AlGaAsSb/AlAsSb, and AlInGaAs/AlInAs, but they require a high number of mirror periods to achieve saturation and the cost of such a growth process would be impractical[35]. A DBR made of Si/SiO_2 is more economical to fabricate, with processes ranging from sputtering to electron beam deposition, but the challenge would now be the transfer of the film from its original growth substrate to the fabricated stack. A transfer process that would preserve the integrity of the film becomes necessary. HCl is known as a selective etchant for InP while leaving InGaAs intact and has been used to isolate InGaAs from sacrificial InP layers with the resulting thin films maintaining an acceptable level of surface roughness and quality[36].

RF magnetron sputtering has been used to deposit a-Si and SiO_2 layers for different applications such as nanolaminate structures[37], superlattices[38], and DBRs for VCSELs[39]. Sputtering utilizes collisions between energized particles of the working gas and the atoms on the cathode target to supply the material to be deposited. Sputtering is done at vacuum pressures to reduce unwanted collisions between gas particles [40].The working gas is commonly argon due to its inertness and availability.

In DC diode sputtering a discharge voltage V_D is applied to ionize the working gas,

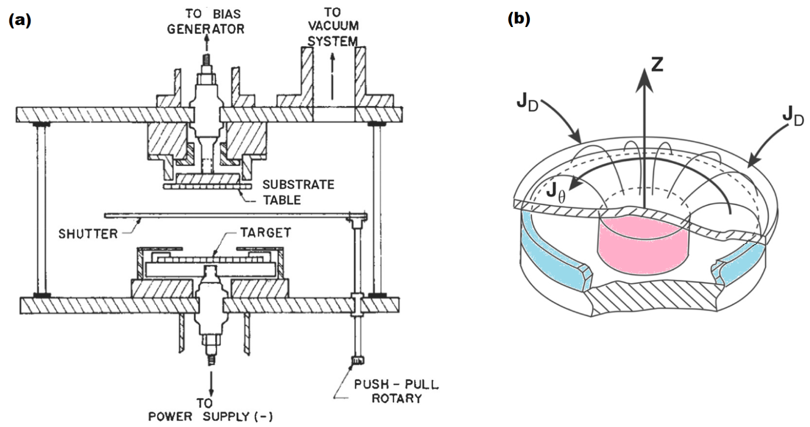

Figure 2.11: Schematics of a magnetron sputtering system (a) with cathode target (b). The plasma is confined by the magnetic field in a toroidal pattern[5].

as shown in Figure 2.11a. The applied V_D must be greater than the breakdown voltage V_B of the gas, in order to ionize the electrons from the gas particles and create plasma via an avalanche effect. DC diode sputtering works well when the target is conductive and metallic, but when the target is insulating due to its inherent properties or impurities then an interfering charge begins to build up. This build-up is avoided by the use of an AC voltage cycled at extremely high or radio frequency (RF). DC sputtering is also limited by the lifespan of the electrons in the vicinity of the target. Adding a static magnetic field $\mathbf{B}$ to confine the ions as shown in Figure 2.11b solves this problem, as it is easier to accelerate a charged ion q with velocity $\mathbf{v}$ according to the Lorentz equation:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{2.9}$$

where $\mathbf{E}$ is the electric field and $\mathbf{F}$ is the force experienced by the particle. Compound films can be formed when a reactive gas such as O_2 is included with the working gas, as a chemical reaction between the reactive gas and sputtered atoms can occur mostly at the surface of the substrate. The whole process is thus called reactive radio frequency magnetron sputtering.

2.4 Electromagnetic Waves in Dielectric Materials

The propagation of electromagnetic waves through dielectric materials can be described using the Fresnel equations, which can be derived from the boundary conditions that constrain the electric and magnetic field components arising as a direct result of Maxwell's equations. Suppose incident light approaches a boundary between two dielectric materials with refractive indices n_1 and n_2 as shown in Figure 2.12. The first case shows s-polarized light, where the electric field is perpendicular to the plane of incidence, while the second case shows p-polarized light, where the electric field is parallel to the plane of incidence. The incident and reflected angles are equal $\theta_i = \theta_r$, and the transmitted angle is given by Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$. The boundary conditions require that the components of the electric and magnetic fields be continuous, thus for s-polarized light, the electric fields are given by $E_i + E_r = E_t$ or $1 + r = t$, where r and t are the reflected and transmitted amplitudes and the incident amplitude is normalized to 1. The y-component of the magnetic field is given by $H_i \cos\theta_i - H_r \cos\theta_r = H_t \cos\theta_t$, which can be rewritten as $n_1 \cos\theta_i(1-r) = n_2 \cos\theta_t t$ using the relation $H = nE/c$. Solving for r gives

$$r_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t} \tag{2.10}$$

while solving for t gives

$$t_s = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}. \tag{2.11}$$

For p-polarized light, the magnetic fields are given by $H_i + H_r = H_t$ or $n_1(1+r) = n_2 t$, thus the electric fields are given by $E_i \cos\theta_i - E_r \cos\theta_r = E_t \cos\theta_t$, which can be rewritten as $n_1(1-r)\cos\theta_i = n_2 t \cos\theta_t$. Solving for r gives

$$r_p = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t} \tag{2.12}$$

while solving for t gives

$$t_p = \frac{2n_1 \cos\theta_i}{n_2 \cos\theta_i + n_1 \cos\theta_t}. \tag{2.13}$$

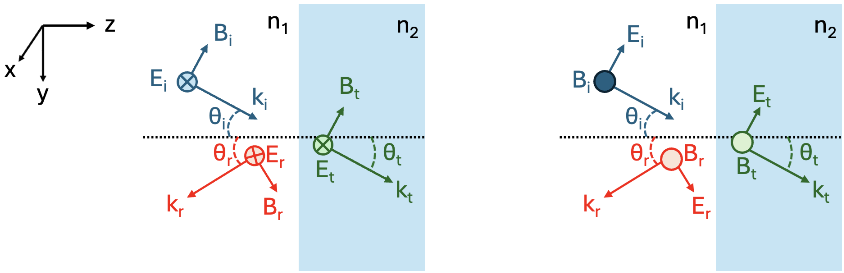

Figure 2.12: Diagram of polarized light at two interfaces. The left image shows the case for s-polarized light, with the electric field pointing into the page. For p-polarized light on the right the magnetic field points outside of the page.

2.5 Transfer Matrix Method

At the boundary between two dielectric materials, the electric field amplitudes of the incident, reflected, and transmitted waves can be related using a transfer matrix. In the schematic shown in Figure 2.13a, a wave with amplitude E_1^+ is moving to the right, and another wave with amplitude E_1^- is moving to the left. For the other side of the boundary, the amplitudes are E_2^+ and E_2^-. The fields can be characterized using two 2×1 matrices, one for each side of the boundary, and a 2×2 interface matrix $\mathbf{D}$ as shown:

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} D_{12} & D_{21} \\ D_{12} & D_{21} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \tag{2.14}$$

The values of the matrix elements D_{ij} can be obtained from inspecting Figure 2.13a. The subscripts i and j refer to the direction of the wave propagation, with t_{12} signifying transmission from medium 1 to medium 2, and r_{12} signifying reflection from medium 1 to medium 2. The fields are thus given by

$$E_2^+ = t_{12}E_1^+ + r_{21}E_2^- \tag{2.15}$$

$$E_1^- = r_{12}E_1^+ + t_{21}E_2^- \tag{2.16}$$

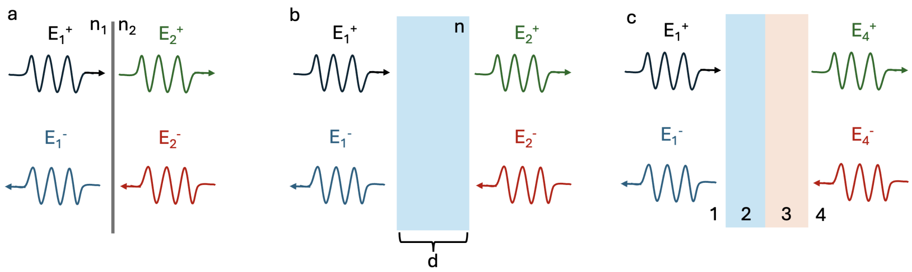

Figure 2.13: Propagation of waves at an interface (a), through a layer with refractive index n and thickness d (b), and through a multilayer system (c).

which can be rearranged to

$$E_1^+ = \frac{E_2^+ - r_{21}E_2^-}{t_{12}} \tag{2.17}$$

$$E_1^- = \frac{r_{12}}{t_{12}}\left(E_2^+ - r_{21}E_2^-\right) + t_{21}E_2^-. \tag{2.18}$$

Considering that the reflection coefficient is $r = (n_1 - n_2)/(n_1 + n_2)$ then $r_{12} = -r_{21}$. For the transmission coefficient $t = 2n_1/(n_1 + n_2)$, and $t_{12} = n_2 t_{21}/n_1$. Substituting these relations into Equation 2.17 and Equation 2.18 with the assumption that $R+T = 1$ yields the matrix equation for the interface $\mathbf{D}$

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \frac{1}{t_{12}} \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix}. \tag{2.19}$$

The propagation matrix $\mathbf{P}$ for a layer of thickness d and refractive index n can be obtained by considering the geometry of the waves in Figure 2.13b, where a layer with refractive index n and thickness d has been inserted between the two waves. The wave moving to the right varies as $E^+ = E_0 e^{-j(k_y y + k_z z - \omega t)}$, and the difference between the two waves at the left and right boundaries is given by $E_2^+ = E_1^+ e^{jkz\cos\theta}$ with the reverse given by $E_2^- = E_1^- e^{-jkz\cos\theta}$. The propagation matrix is thus given by

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} e^{jkd\cos\theta} & 0 \\ 0 & e^{-jkd\cos\theta} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \tag{2.20}$$

where $k = 2\pi n/\lambda$ is the wavenumber in the medium. This analysis can be extended to a two layer system by multiplying the interface and propagation matrices while considering the order of the layers. Starting from the rightmost boundary as shown in Figure 2.13c, the fields on both sides are related using the interface matrix by $\mathbf{E}_{3R} = \mathbf{D}_{34}\mathbf{E}_4$, where $3R$ is the field at the right side of layer 3. The fields at the left side of layer 3 are related to the right side by the propagation matrix $\mathbf{E}_{3L} = \mathbf{P}_3\mathbf{E}_{3R}$, which in turn can be related using another interface matrix to the field across the relevant boundary. This results in the following matrix equation relating the fields at the leftmost region to the rightmost region:

$$\mathbf{M} = \mathbf{D}_{12}\mathbf{P}_2\mathbf{D}_{23}\mathbf{P}_3\mathbf{D}_{34} \tag{2.21}$$

such that

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix}. \tag{2.22}$$

This method allows the analysis of any multilayer system using matrix multiplication of the component interface $\mathbf{D}$ and propagation $\mathbf{P}$ matrices. The reflection coefficient r for a multilayer structure is simply

$$r = \frac{M_{21}}{M_{11}} \tag{2.23}$$

since $t = 1/M_{11}$ and $r = M_{21}/M_{11}$.

2.6 Envelope method for refractive index

CHAPTER 3

METHODOLOGY

3.1 Initial growth of a-Si and SiO_2 thin films via electron beam evaporation

Glass slides and Si wafers were used as substrates for the initial growth via electron beam evaporation of a-Si and SiO_2 thin films respectively. The substrates were cleaned by sonication in acetone, isopropanol, and deionized water for 5 minutes each, after which they were blow-dried with nitrogen gas. The substrates were then loaded into the ULVAC electron beam evaporation chamber as shown in Figure 3.1, where a base pressure of $4.5 \times 10^{-4}\,\mathrm{Pa}$ was achieved before actual deposition. Deposition was carried out at room temperature, and the nominal deposition rates and thicknesses of the films were monitored using a quartz crystal microbalance (QCM). After unloading the samples from the chamber, the thicknesses of the films were measured using x-ray reflectivity (XRR) with a Malvern Panalytical Empyrean as shown in Figure 3.1. The refractive index of the a-Si film was determined using Swanepoel's method via UV-Vis spectroscopy with a Shimadzu UV-Vis spectrophotometer shown in Figure 3.1 while the refractive index of the SiO_2 film was measured using a reflectance spectroscopy setup as shown in Figure 3.1. The measured refractive indices of the a-Si and SiO_2 films were used for the design of the a-Si/SiO_2 ARC stack.

3.2 Simulation and growth of a-Si/SiO_2 antireflective coatings

An implementation of the transfer matrix method in Python was used to calculate the optimal thicknesses of the Si and SiO_2 layers for the antireflective coating stack. The refractive indices of the a-Si and SiO_2 layers were obtained from the previous section. The optimal thicknesses of the layers were calculated to minimize the reflectance at $1550\,\mathrm{nm}$ and $780\,\mathrm{nm}$, which are the wavelengths of the pump and probe beams which

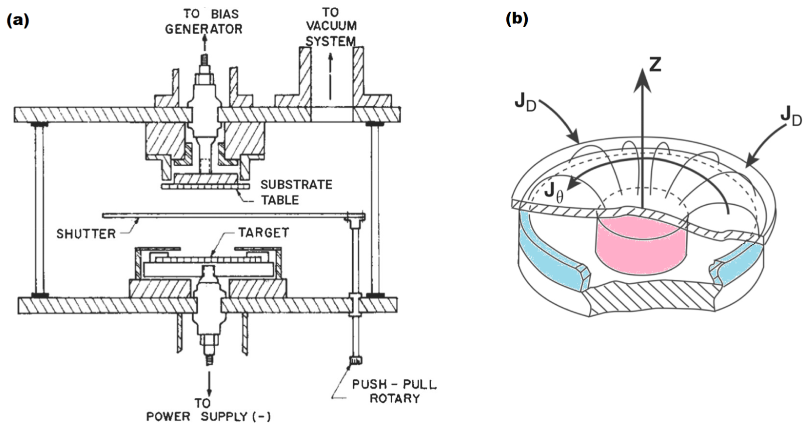

Figure 3.1: Schematics of a magnetron sputtering system (a) with cathode target (b). The plasma is confined by the magnetic field in a toroidal pattern[5].

can be used in the THz-TDS spectroscopy system. The program iterated over a range of thicknesses for each layer while calculating the reflectance of the stack, selecting the thicknesses that resulted in the lowest reflectance at the target wavelengths. The thicknesses obtained from the simulation were then used to grow the a-Si/SiO_2 antireflective coating on Si wafers and the InGaAs sample via electron beam evaporation. The reflectance of the samples with the antireflective coating was measured via reflectance spectroscopy and compared to the simulated reflectance to verify the accuracy of the simulation and growth process.

3.3 THz TDS of InGaAs on a-Si/SiO_2 DBR

The THz emission characteristics of the InGaAs on a-Si/SiO_2 DBR PCA will be measured with a standard THz-TDS spectroscopy system. The beam from a $1.55\,\mu m$ femtosecond laser will be split into pump and probe beams using a beam splitter, with the pump beam exciting the InGaAs on a-Si/SiO_2 DBR PCA and the probe beam optically gating a THz detector. The time domain and representative frequency domain spectra of the InGaAs on a-Si/SiO_2 DBR PCA will be analyzed.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Thicknesses and refractive indices of a-Si and SiO_2 thin films

Figure 2.1 and Figure 4.1 show the XRR measurements of various a-Si films grown on glass substrates with varying thicknesses according to the quartz crystal microbalance (QCM) readings. The measured thicknesses of the a-Si film based on XRR are summarized in Table 4.1, while for SiO_2 the results are summarized in Table 4.2. Linear regression of the measured thicknesses versus the QCM readings for a-Si as shown in Figure 4.1 yields the equation $y = 167.82x + 2.749$ with $R^2 = 0.9999$, where y is the measured XRR thickness in nanometers and x is the QCM reading in kiloangstroms. Similarly, linear regression of the measured XRR thickness and the QCM values for SiO_2 as shown in Figure 4.1 yields the equation $y = 199.49x - 5.4845$ with $R^2 = 0.9937$.

The results of the regression analysis indicate that the QCM readings during deposition scale linearly with the actual thicknesses of the films as measured by XRR. The high R^2 values suggest that the QCM is a reliable tool for monitoring film thickness during deposition, and the linear relationships can be used to calibrate the QCM readings for future depositions. The lower R^2 value in the regression for SiO_2 is attributed to the variation in the deposition rate during the growth of the SiO_2 films, which is

QCM (kA)	XRR (nm)
0.035	6.9
0.138	27.5
0.104	21.1
0.069	13.7
1.52	257.7

Table 4.1: a-Si thickness

QCM (kA)	XRR (nm)
1.05	204.3
0.525	103.7
0.884	162.9
1.369	270.8

Table 4.2: SiO2 thickness

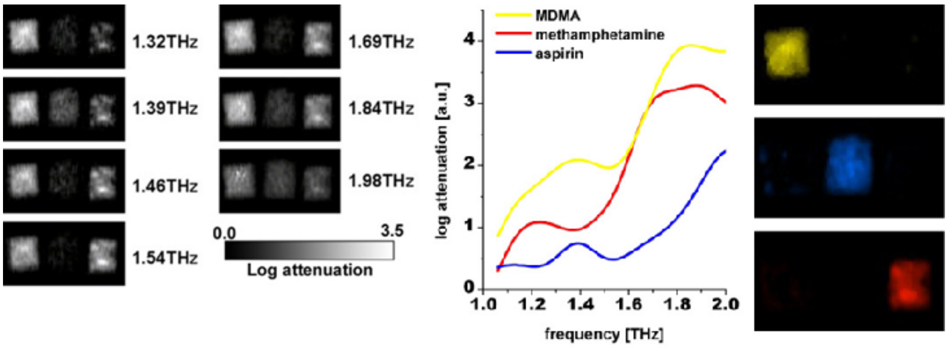

Figure 4.1: THz identification of MDMA, methamphetamine, and aspirin through a paper envelope. The substances are differentiated using their THz spectra

caused by the difficulty in maintaining a stable deposition rate for SiO_2 due to its low thermal conductivity and tendency to sublime instead of melting into a uniform liquid pool during heating by the electron beam as is the case for other materials˙ Figure 4.1 and Figure 4.1 show the source materials inside the crucibles for electron beam evaporation, with the a-Si source material melting uniformly while the SiO_2 source material shows localized heating and uneven melting.

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Gantt chart

ACTIVITIES	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER
Procure required supplies for fabrication									
Simulation of DBR parameters									
Growth of a-Si/SiO2 DBR via reactive RF magnetron sputtering									
Transfer of InGaAs photoconductive layer to a-Si/SiO2 DBR stack									
THz time domain spectroscopy of InGaAs PCA integrated with a-Si/SiO2 DBR									
Data collection and analysis									
Thesis writing									

ITEM	DESCRIPTION	UNIT COST	QUANTITY	TOTAL
Fabrication supplies				
Acetone	4L (JTBaker)	2,300.00	1 bottle	2,300.00
38% Hydrochloric acid	2.5L (JTBaker)	3,955.00	1 bottle	3,955.00
Photoresist	250ml (ma-P 1225)	37,200.00	1 bottle	37,200.00
Developer	2.5L (mr-D 526/S)	24,100.00	1 bottle	24,100.00
Si wafers	Si(100), 2", undoped	11,342.50	2 pcs	22,685.00
Others				
Printing				5,000.00
Characterization	Scanning electron microscopy			10,000.00
Total				**105,240.00**